

Tea consumption didn't modify the risk of fracture: a dose-response meta-analysis of observational studies

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Background Fractures are important causes of healthy damage and economic loss nowadays. The conclusions of observational studies on tea consumption and fracture risk are still inconsistent. The objective of this meta-analysis is to determine the effect of tea drinking on the risk of fractures. Methods A comprehensive literature search was conducted in PubMed, Embase and reference lists of the relevant articles. Observational studies that reported an estimate of the association between tea drinking and incidence of fractures were included. A meta-analysis was conducted by the STATA software. Results A total of 9 studies involving 147,950 individuals that examined the association between tea consumption and risk of fractures were included in this meta-analysis. The odds risks (ORs) with 95% confidence intervals (CIs) were pooled using a random-effects model. The pooled OR of 9 observational studies for the tea consumption on risk of fracture was 0.89 (95% CI, 0.78-1.04). In the subgroup analyses, no significant association was detected in neither cohort studies ($n = 3$; OR, 0.97; 95% CI, 0.89-1.06) nor case-control studies ($n = 6$; OR, 0.91; 95% CI, 0.70-1.19), respectively. No significant association was detected in the dose-response meta-analysis. Conclusions Tea consumption might not be associated with the risk of fractures. The following large-sample and well-designed studies are required to confirm the existing conclusions. Virtual slides The virtual slide(s) for this article can be found here: <http://www.diagnosticpathology.diagnomx.eu/vs/5309904231178427>.